

PEDRO PALACIOS ALMENDROS

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EDUCATION

- Ph.D. in Computer Science**, *University of Illinois at Urbana-Champaign (UIUC)* 2025 - current
Advisor: Josep Torrellas. GPA: 4.0 out of 4.0.
Research topics: computer architecture, ML, and distributed systems of heterogeneous accelerators.
Relevant coursework: Parallel Computer Architecture, Machine Learning Compilers, Cloud Storage Systems, Deep Generative Models.
- M.Sc. in Computer Science**, *École Polytechnique Fédérale de Lausanne (EPFL)* 2023 - 2025
GPA: 5.96 out of 6. Ranked 1st in Computer Science department and 2nd university-wide.
Relevant coursework: Advanced multiprocessor architecture, Advanced computer architecture, Distributed algorithms, Advanced compiler construction, Systems for data management, VLSI.
- B.Sc. in Computer Science (240 ECTS)**, *Universidad Complutense de Madrid (UCM)* 2018 - 2023
GPA: 9.89 out of 10. Ranked 1st in graduating class. Part of a double degree program.
- B.Sc. in Mathematics (240 ECTS)**, *Universidad Complutense de Madrid (UCM)* 2018 - 2023
GPA: 9.76 out of 10. Ranked 1st in graduating class. Part of a double degree program.
- Exchange program during bachelor**, *University of California, San Diego (UCSD)* 2021 - 2022
GPA: 4.0 out of 4.0. Awarded provost honors in all three terms.

INDUSTRY EXPERIENCE

- Amazon Web Services (AWS) – Annapurna Labs**, *Applied Scientist Intern* May 2026 - August 2026
Developed a fine-grained power model for AWS Trainium ML accelerators and worked on compiler-hardware power co-optimization techniques
- IMEC**, *Research Intern* January 2025 - June 2025
Designed and evaluated a near-KV cache Compute-near-Memory accelerator using emerging CCD 3D memory for mobile LLM inference.

ACADEMIC EXPERIENCE

- i-acoma group at UIUC**, *Research Assistant* August 2025 - current
Working with Prof. Josep Torrellas on computer architecture for ML, and distributed systems of heterogeneous accelerators within the SRC JUMP 2.0 ACE Center for Evolvable Computing.
- Embedded Systems Laboratory at EPFL**, *Research Assistant* June 2024 - January 2025
Worked on the design, programming, Gem5 simulation and integration into a RISC-V SoC of a low-power mixed-precision ML accelerator supporting structured pruning.
- Embedded Systems Laboratory at EPFL**, *Research Assistant* March 2024 - June 2024
Worked in a joint research project with IMEC on micro-architectural simulation, evaluation and optimization of GPUs for 3D stacked transistor technologies.
- Dynamically Reconfigurable Hardware Group at UCM**, *Research Assistant* 2022 - 2023
Developed custom hardware accelerators for real-time hyperspectral image processing on FPGAs.

PUBLICATIONS

- [P. Palacios](#), B. Wang, F. Garcia, G. Ansaloni, D. Atienza, J. Klein, D. Biswas, *CCD-KV: Exploiting High-Density 3D Charge-Coupled Device Memory for Efficient On-Device LLM KV Caches*, in ASPDAC 2027.
- G. Ajmone, [P. Palacios](#), C. Sung, D. Kroening, T. Bourgeat, J. Choi, *Dynamic Race Detection for HW/SW Co-Designed Systems* submitted to ASPLOS 2027.
- [P. Palacios](#), R. Medina, J.L. Rouas, G. Ansaloni and D. Atienza, *Systolic Arrays and Structured Pruning Co-design for Efficient Transformers in Edge Systems*, in GLSVLSI 2025. [Link](#).

- J.L. Rouas, C. Brazier, L. B. Letaifa, R. Medina, P. Palacios, D. Atienza, G. Ansaloni, *Structured Pruning for Efficient Systolic Array Accelerated Cascade Speech-to-Text Translation*, in Interspeech 2025. [Link](#).
- P. Palacios, D. Báscones, C. González and D. Mozos, *A Real-Time FPGA Implementation of the LCMV Algorithm for Target Classification in Hyperspectral Images Using LDL Decomposition*, in IEEE Transactions on Geoscience and Remote Sensing, 2024. [Link](#).

PATENTS

- P. Palacios, B. Wang, J. Klein, F. Garcia, G. Ansaloni, D. Atienza, *Transformer Auto-Regressive Attention Accelerator Using 3D String-Memory Key-Value Cache*, Patent application in progress, 2026.

WORKSHOP PAPERS

- C. Block, P. Palacios, A. Farrell, G. Gerogiannis, J. Torrellas, *Performance-Driven Composite Prefetching with Bandits* in DPC4 at HPCA 2026.
- P. Palacios, J. Klein, A. Sharma, D. Abdi, G. Ansaloni, F. Garcia, D. Biswas, D. Atienza, J. Myers, *Full Stack Framework for Architectural Exploration of IGZO-Based Compute-near-DRAM* in IMEC PTW 2025.
- P. Palacios, R. Medina, G. Ansaloni, D. Atienza, *HEEPstor: an Open-Hardware Co-design Framework for Quantized Machine Learning at the Edge* in Open-Source Hardware Workshop at ACM CF 2025.

PRIZES AND AWARDS

- Awarded the “*Andrew & Shana Laursen Fellowship*” for top incoming UIUC grad students (\$30,000) 2025
- Awarded the “*2nd Best Master Academic Record*” university-wide across all departments by EPFL 2025
- Awarded the extremely competitive “*La Caixa*” *Postgraduate Abroad Fellowship* (\$65,000) 2023-2025
- Received the *Award for the Best Academic Record* by Universidad Complutense de Madrid 2023
- Awarded the *Excellence Scholarship* four times by the Community of Madrid (\$10,000) 2019-2023
- Participated in and achieved various positions in several competitive programming events, and hardware competitions, as well as national mathematics, computer science, and physics olympiads 2016-2023

OTHER PROJECTS

- **Cloud FPGA accelerator orchestration.** Working with Jovan Stojkovic and Josep Torrellas on dynamic management and orchestration of heterogeneous accelerator swarms in FPGA-enabled datacenters.
- **BNN SIMDRAM compiler:** End-to-end MLIR compiler for mapping 1-bit neural networks to the SIMDRAM PIM architecture, including custom dialect, passes, runtime, and Gem5 evaluation. [Link](#).
- **Heterogeneous multi-chiplet accelerator exploration.** Developing an SST-based simulation and design-space exploration framework for multi-chiplet accelerators, including ML workload mapping to the accelerators.
- **Master’s Thesis:** *System-Level Exploration of Emerging Memories for Compute-near-Memory*. Developed a modeling framework for emerging IGZO eDRAM and CCD 3D memories, spanning from device-level simulations to architectural simulation. Designed a CnM architecture to map LLM inference to emerging 3D CCD memory.
- **Asgard:** Hardware-partitioning for terabyte-scale bare-metal cloud servers using the Midgard address space that improves core assignment granularity and reduces LLC communication, keeping high associativity. [Link](#).
- **RISC-V SoC Implementation in 28nm FDSOI:** Integrated the open-source SweRV RISC-V core into a custom SoC, and developed TCL scripts for FDSOI 28nm synthesis, floorplanning, and place-and-route.

TOOLS AND SKILLS

C++, Rust, Java, Scala, GoLang, CUDA, Docker, Python, Bash, VHDL, SystemVerilog, Chisel, RISC-V, Gem5, AccelSim, Ramulator, DRAMPower, QEMU, Vitis HLS, Vivado, Cadence Genus, Cadence Innovus, perf, Valgrind, Linux Driver Development, LLVM, MLIR, XLA, Triton, Amazon Trainium NKI, AMD MLIR-AIE, PyTorch, JAX.